PHYSICS

Paper 5054/02
Structured and Free Response

General comments

The Examiners are of the opinion that the performance the candidates in this session, was very much in line with that which has been achieved in the most recent examination sessions. The overwhelming majority of the candidates answered some questions correctly whilst making errors in others. Consequently, they finally obtained a mark somewhere in the middle of the range. The actual mark obtained was, of course, an indication of the breadth and depth of the understanding of the particular topics which this year's questions examined and of the candidate's ability to express this on paper. It did, however, seem that rather fewer candidates than usual scored the very highest marks possible. This, on the other hand, seems to have been more than balanced by a slight but noticeable and significant rise in the numbers who produced performances which were very creditable and decent but which did not quite reach the highest standards throughout the paper. At the other end of the mark range, there was a further decrease in the fraction of candidates who scored very few marks at all and many of the candidates whose scores did fit into this band of marks, left significant sections of the question paper blank and only attempted a few parts of the questions in ***Section B***. There was a slight increase in the number of candidates who infringed the paper rubric by answering three questions in ***Section B***. In this session, a further opportunity for this arose in **Question 8** where the **OR** section tested material from ***Section 25*** of the syllabus which is, of course, optional. A very small (but larger than usual) number of candidates omitted to answer **Question 8** at all and as a consequence scored none of the marks available. It is possible that some of these were unable to decide which section to answer and then forgot to return to the question at the end of the examination.

Almost all candidates produced answers which indicated that the question had been properly understood and the answers given revealed that the vast majority of candidates had achieved a fluency in written English which was more than sufficient for the demands of this paper. There were probably fewer opportunities than usual for writing significant amounts of continuous prose and, where the chance did arise, it was taken by the candidates who, in the main, wrote well and accurately even when the physics itself had been poorly understood. It should be borne in mind by the candidates that in an examination such as this, clarity of expression is important and answers where the meaning is clear will be capable of achieving all the marks allocated to the question. Diagrams, bullet points and drawings can all help where a verbal description might prove laboured and less transparent. This applies to all the questions in the paper even where only answer lines are supplied on the paper. Although some attempt is made to match the number of printed answer lines in ***Section A*** to the number of marks available in that section, candidates should not feel that, for a given question or question section, they are expected to fill up the lined answer sections completely.

Very occasionally, the handwriting of a candidate is so indistinct that the Examiner is unable to award the marks available despite having made a serious attempt to understand what was intended. This situation is most likely to arise where the candidate is using an unusual word or phrase which the Examiner might not expect to encounter in the answer to the particular question.

As is often the case, the numerical calculations proved a fecund source of marks for the better candidates who may, in some questions, score 2 or even 3 marks by producing a few lines of accurate calculation concluding with the correct answer and the correct SI unit. There are, of course, a few dimensionless quantities where to include a unit would be erroneous; refractive index is an example of such a quantity. Where a correct answer is not supplied by the candidate, a small number of marks is usually available for showing the appropriate working or sometimes for just quoting the appropriate formula and so the better candidates invariably show the stages by which the final answer is achieved. Other candidates might easily forfeit marks by not doing so.

In **Question 11(a)(iii)** some candidates used a calculator set in the radian mode when calculating the refractive index of glass from angles given in degrees; this gives the wrong answer. It was especially unfortunate when this was coupled with a failure to show any working and three marks were lost as a result. Candidates who give contradictory answers to the same question cannot be credited for stating the correct

answer. An example arose in **Question 11(b)(iii)** where the surprisingly large number of candidates who stated that the image was both magnified and diminished scored no marks.

There is a tendency for certain questions to produce answers which are too general and do not focus sufficiently on the specific point being examined; candidates should be strongly advised to read the question carefully and to restrict answers to remarks which answer the very specific point being asked. For example: **Question 2(a)(i)** was about measuring the volume not the density; **Question 10(a)** is concerned with electrical conduction not heat conduction. The syllabus outlines the meaning of terms such as: *state, list, determine* and *show.* Reference to this list should help candidates to give a response of the sort which the examination is expecting.

Comments on specific questions

Section A

Question 1

(a) This section asked what happened at the two given times and most candidates restricted themselves to the two decelerations or to their causes. It was surprising that several candidates referred to reaching terminal velocity or to uniform or even gradual decelerations.

(b) This part of the question was generally well answered. Marks were usually lost when candidates who gave the correct answer continued and supplied further, inaccurate details. A common response had the parachutist accelerating at an increasing rate and the decrease in the acceleration was often inaccurately described as a deceleration.

(c) A disappointingly small number of candidates scored both marks. The forces were quite commonly correctly named but the fact that the resultant force on the parachutist was zero was rarely stated with sufficient clarity. Sometimes the forces were simply equal, sometimes just opposing and some candidates referred to the resultant force being cancelled out by another force.

(d) This question was very commonly correct but some candidates used a speed of 4.5 m/s or misread the graph in other ways. Likewise, marks were lost by candidates who rearranged the formula incorrectly or gave the unit of distance as m/s.

Answers: ***(d)*** *150 m*

Question 2

(a) **(i)** This part was very well answered since most candidates were thoroughly familiar with the displacement method for determining volume. A few candidates omitted to mention the use any liquid in the measuring cylinder. The less precise answers did not state that the level of the liquid was recorded at any stage and sometimes it was not clear which two values were being subtracted. The spelling: *substracted* was very widespread. In **(ii)**, the weaker answers did not explain why the limited volume of an ordinary laboratory measuring cylinder was relevant in this case. Some candidates stated that the mass of the rock was too great or that it would break the glass.

(b) The density of rock A was correctly calculated very frequently indeed. Only very occasionally was the wrong formula used or an incorrect final unit given. Some candidates gave the correct answer in kg/m^3 and scored full marks.

(c) The majority of candidates were correct to state that rock C was made from a different material and correctly explained how this was deduced. The most common error, however, was to give rock D as the answer because of its much greater mass and volume.

Answer: ***(c)*** *4.59 g/cm^3*

Question 3

(a) **(i)** Whilst some candidates did name the renewable energy source, a very large number did not and a variety of inspired or random guesses was very evident. Many candidates had some idea of the point which **(ii)** was testing. Many others, however, used phrases such as: *can be used again, can be recycled.* The word *renewable* was given in the question and cannot be used to explain its own meaning.

(b) These two calculations were performed correctly by many candidates but some answers were spoiled by incorrect final units. The correct and straightforward J was very common but some used the units of the quantities given in the question or made errors when trying to deduce the correct unit from the formula. Candidates who used the formula in the form: $L_v = ml_v$ were sometimes uncertain as to which quantity represents the *specific* latent heat and errors of substitution arose as a result.

Answer: ***(b)(i)*** *3.36×10^8 J* ***(ii)*** *2.3×10^8 J*

Question 4

(a) The good absorption property of black surfaces was widely given but several candidates spoiled the answer and lost a mark by also referring to their good emission property which does not constitute an accurate explanation in this case. Very few candidates made any reference to *radiation* or *IR radiation* and so the second mark was only rarely awarded.

(b) Many candidates misunderstood this question. Answers tended to explain why it was desirable for the hot water to be in the tank rather than explaining how it got there. Very few answers made any reference to the reduced density of hot water or to the setting up of a convection current; this part was poorly answered.

(c) **(i)** was generally well answered although some candidates referred to importance of electrical insulation. Many materials in **(ii)** were sensible and appropriate but some candidates suggested double-glazing the tank or manufacturing it from copper or painting it white.

Question 5

(a) Most candidates realised that the conduction of heat was due to the atomic vibrations and scored the first mark in **(i)** but some merely referred to the vibration instead of the energy being passed on. Only a correct reference to the transmission of energy scores the second mark. In **(ii)**, many candidates talked about faster vibrations or the expansion of atoms. The mark was awarded for vibrations of greater amplitude.

(b) Most candidates were able to obtain one mark here by giving one relevant change either in the arrangement or in the motion of the atoms. A significantly smaller number, however, was able to get both marks. Answers which referred only to the macroscopic properties of liquids and solids did not answer the question and did not score.

Question 6

(a) The vibration of the cone or of the air molecules near to it was mentioned by many candidates and so the first mark was awarded very commonly indeed. Very few candidates stated that the vibration of the air molecules was longitudinal and when it was scored, the second mark was usually credited to candidates who referred to the compressions and rarefactions produced in the air.

(b) Although not universally so, the highest frequency that a human with normal can hear was accurately quoted very widely. Likewise, the calculation in **part (ii)** was very commonly correct. Some candidates, however, used the answer to **part (i)** and not the frequency value given in the question to calculate the **shortest** wavelength of audible sound and not the **longest** as requested.

Answer: ***(b)(i)*** *18 000 – 20 000 Hz* ***(ii)*** *17 m*

Question 7

(a) A large number of candidates wrote down the correct answer steel in **(i)** but it is clear that many others do not realise the significance of the words *soft* and *hard* when applied to ferromagnetic materials. There was a similar confusion in **(b)(i)**. Whilst some candidates answered **(ii)** well, others did not mention the need for a current in the coil or at least for the switch to be closed.

(b) This part of the paper was very poorly answered. Most candidates drew diagrams which suggested a very limited understanding of this topic. Unfortunate features of many such diagrams included: field lines not linking up, field lines bowing outwards not inwards, field lines being concentrated on the component, field lines crossing and touching.

Question 8 EITHER

The question was well understood by many who attempted it and 4/4 was quite a common mark. Marks were forfeited for answers where the amplitude or the periodic time were out by a factor of 2 or where the diagram was so poorly drawn that it was not clear which length should be taken to represent the amplitude or the periodic time.

Question 8 OR

(a) This part of the question was almost universally very poorly answered and it is clear that almost none of the candidates has any idea of the manner in which a transistor functions in a circuit such as this.

(b) Very few candidates suggested a suitable use for the moisture detector; several candidates stated that it could be used to switch on a pump or it could measure moisture levels. The device referred to in the question could do neither of these.

Section B

Question 9

(a) The calculations enabled many of the better candidates to score well here. Familiar errors in the unit of acceleration and the inaccurate rounding to an appropriate number of significant figures were responsible in some cases for candidates' not scoring full marks on **(i)** and **(ii)**. When applying $F = ma$ some candidates used a = 10 m/s^2 rather than the calculated value from **(i)**. **Part (iii)** was much more poorly answered. Most candidates relied on vague and unconvincing answers involving the application of opposing forces by the airbag or to air escaping from the bag and hence absorbing the force in some uncertain manner. Few of those who stated that the deceleration was reduced realised that this was because of the increase in the duration of the deceleration.

(b) Many candidates were able to define pressure but some who might have scored the mark gave sloppy and imprecise definitions such as: *the force acting on a certain area.* Many candidates simply stated the defining equation and scored the mark. In **(ii)** it was surprisingly rare for a candidate to score both marks. Again several candidates reverted to answers which repeated those already given for **(a)(iii)**.

(c) Again many of the higher scoring candidates were able to obtain the 3 marks in **(i)** for just a few lines of accurate calculation and the correct answer with the correct unit. Given the very specific wording of the question, an unfortunately large number of candidates used the volume 29 400 cm^3 in the calculation. Most candidates realised that pressure was caused by molecular collisions with the container walls but many of the candidates who did not score full marks here, wrote about the airbag alone and did not make a specific statement concerning the reduction in the number of air molecules in the cylinder.

Answer: ***(a)(i)*** *4.7 m/s^2* ***(ii)*** *23 N* ***(c)(i)*** *280 000 Pa*

Question 10

(a) Most candidates realised what was required here and were able to score quite well on this section. Answers which included a circuit diagram tended to score well and helped candidates to make clear their intentions. There was a significant number of answers involving heat conduction and some tested for good thermal conductors before stating that good thermal conductors are good electrical conductors. Such answers were not credited; nor were experiments which involved the administering of electric shocks to human beings. Almost all candidates were able to name a good and a poor electrical conductor.

(b) This part of the question was, in general, quite poorly answered. Although most candidates were able to define resistance, many restricted their answers to general statements about resistance being a measure of the opposition to current flow or the force which opposes current flow and did not give its exact numerical definition. It was noticeable that in **part (ii)**, many answers stated how the current varied with potential difference or referred to the resistance being directly proportional to the potential difference. The first does not answer the question and the second is neither true nor readily deduced from the sketch graph in the question. Many candidates used expressions such as: *P gets smaller as the potential difference rises*. It is not clear what P means here as in the question it is stated that P is a component not a measurement or quantity. Whilst many candidates did state that component P is a tungsten filament lamp or made one or two other equally probable suggestions, a rather larger number produced answers such as: *resistor, thermistor,* and *diode.* In **(iv)** only a very small number of candidates realised that the effect was caused by the rising temperature of component P and of these few scored more than 1/2.

(c) Many of the better candidates were again able to score well on the calculations. Common errors included the use of inaccurate rearrangements of the Ohm law: $V = I/R$. Some candidates halved the potential difference calculated in **(ii)** for the answer in **(iii)** whereas the values are the same. The value of R in **part (iv)** was often given as a fraction; this was not accepted.

Answer: ***(c)(i)*** *1.0 A, 1.0 A* ***(ii)*** *8.0 V* ***(iii)*** *8.0 V* ***(iv)*** *13 Ω*

Question 11

(a) **Part (i)** of the question was often well answered and even candidates in the middle of the mark range often scored 2/2. In **(ii)**, some knew the three correct answers whilst others were obliged to guess and sometimes gave answers which were inconsistent with each other: the velocity is reduced whilst the wavelength and frequency increase. **Part (iii)** was frequently well answered and there was little indication to suggest that candidates, who know that the value is 1.5, simply wrote it down without calculation.

(b) In **(i)** there was a fair amount of uncertainty amongst the candidates as to what was being asked for. Many gave an answer which suggested that they were trying to define the focal point and those candidates who referred to it by name used terms such as: *the focal, the focus, the principal length.* In diagrams, the diameter of the lens was frequently labelled: *the principal axis.* The diagrams were usually fairly untidy and arrows drawn to indicate the focal length would often begin and end short of the correct positions. In such cases, the second mark was forfeited. It was interesting that some weaker candidates were able to produce large, carefully drawn and accurate diagrams in **part (ii)** and hence score some useful marks. The answers given here were variable in quality but most candidates who attempted this part of the question had some idea of what was required. A common error was to position the object at some random location rather than where the scale located it and thus to obtain an image whose magnification lay outside the allowable limits. Some candidates marked the point where the two refracted rays crossed as the image and did not link this point to the principal axis to produce an extended image; they scored only 2/3. Many of the candidates were able to obtain at least one and frequently both of the marks available in **part (iii)**. This was sometimes as the result of an error in the diagram being carried forward.

Answer: ***(a)(iii)*** *1.52*